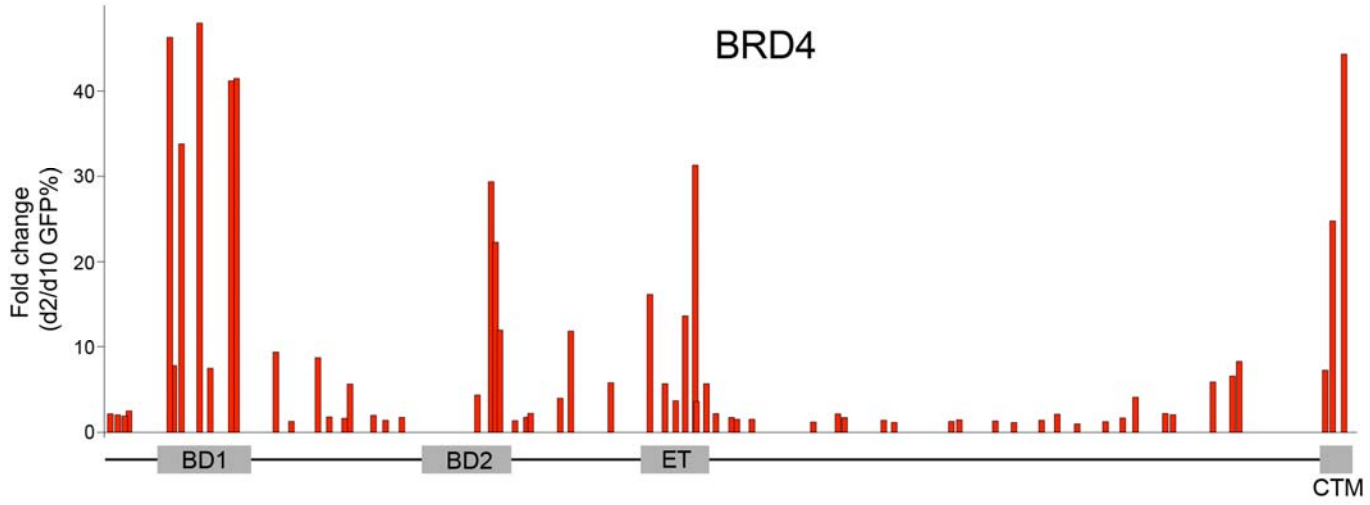
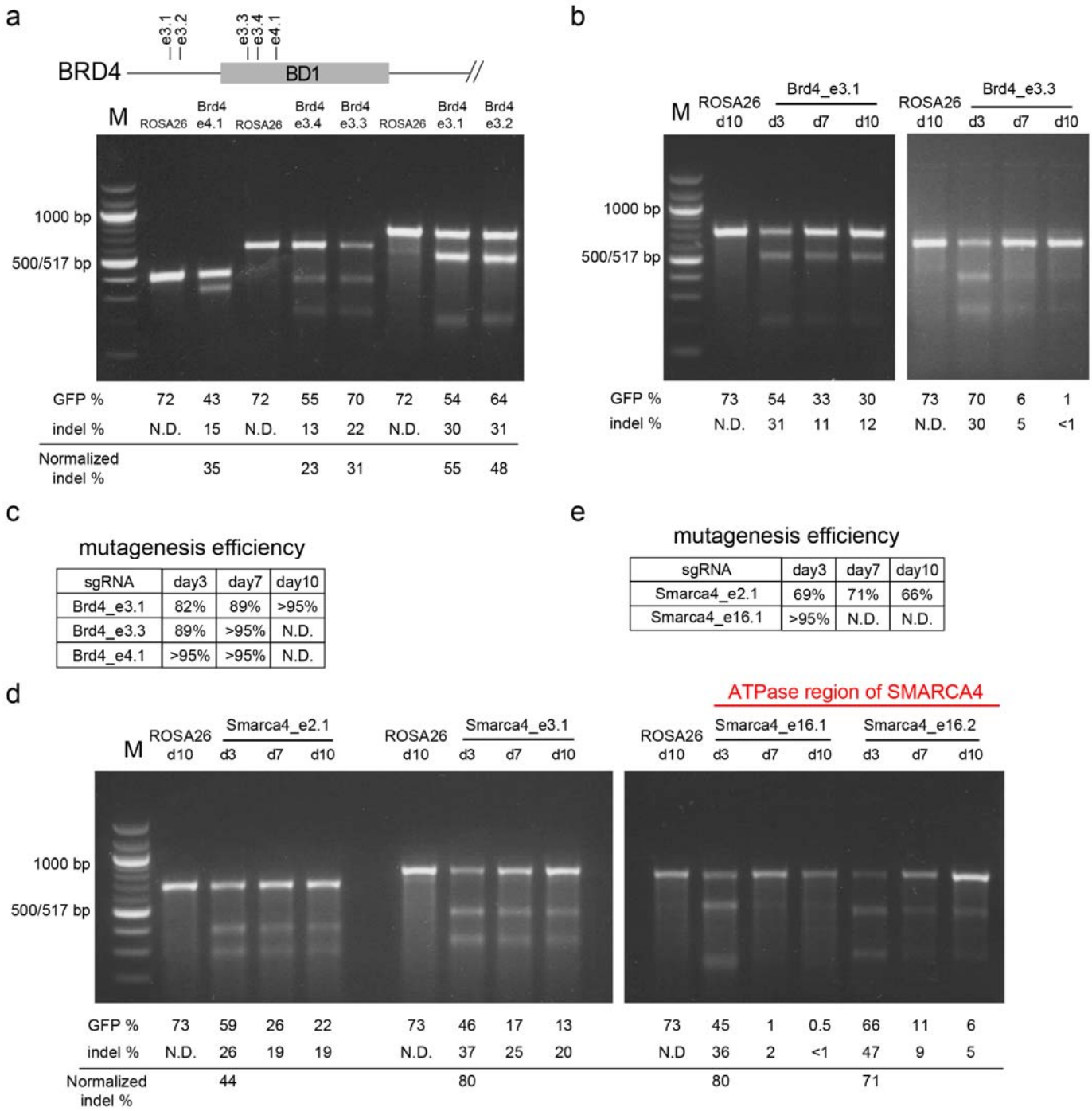


Supplementary Figure 1

Negative selection analysis of sgRNAs targeting all *Brd4* exons comparing day 2 to day 10 time points.

Systematic evaluation of 64 *Brd4* sgRNAs in negative selection experiments, targeting each *Brd4* exon. The location of each sgRNA relative to the Brd4 protein is indicated along the x-axis. BD1: bromodomain 1, BD2: bromodomain 2, ET: extra-terminal domain, CTM: C-terminal motif. Plotted is the fold change of GFP positivity comparing day 2 and day 10 post-infection, representing the average of three independent biological replicates.



sgRNA	day3	day7	day10
Brd4_e3.1	82%	89%	>95%
Brd4_e3.3	89%	>95%	N.D.
Brd4_e4.1	>95%	>95%	N.D.

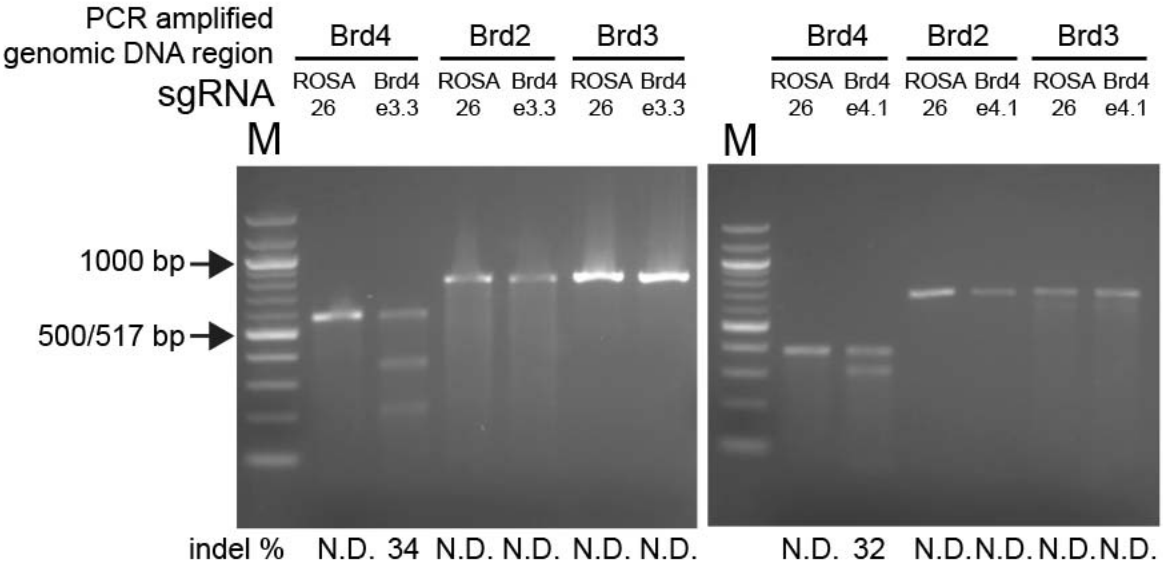
sgRNA	day3	day7	day10
Smarca4_e2.1	69%	71%	66%
Smarca4_e16.1	>95%	N.D.	N.D.

Supplementary Figure 2

SURVEYOR assay and deep sequencing analysis of indel mutations induced by various *Brd4* or *Smarca4* sgRNAs.

(**a**) Top panel, location of Brd4 sgRNAs relative to the domain architecture of the Brd4 N-terminus. Bottom panel, SURVEYOR assay of indel mutations at corresponding Brd4 genomic DNA regions. Analysis was performed at day 3 post-infection. sgRNA targeting ROSA26 locus serves as the negative control. The GFP+/sgRNA+ percentages of each sample at day 3 are labeled under the gel image. The indel% was calculated based on the relative intensity of the DNA bands using ImageJ software. The normalized indel% was calculated by dividing the indel% by the GFP%. Representative image of

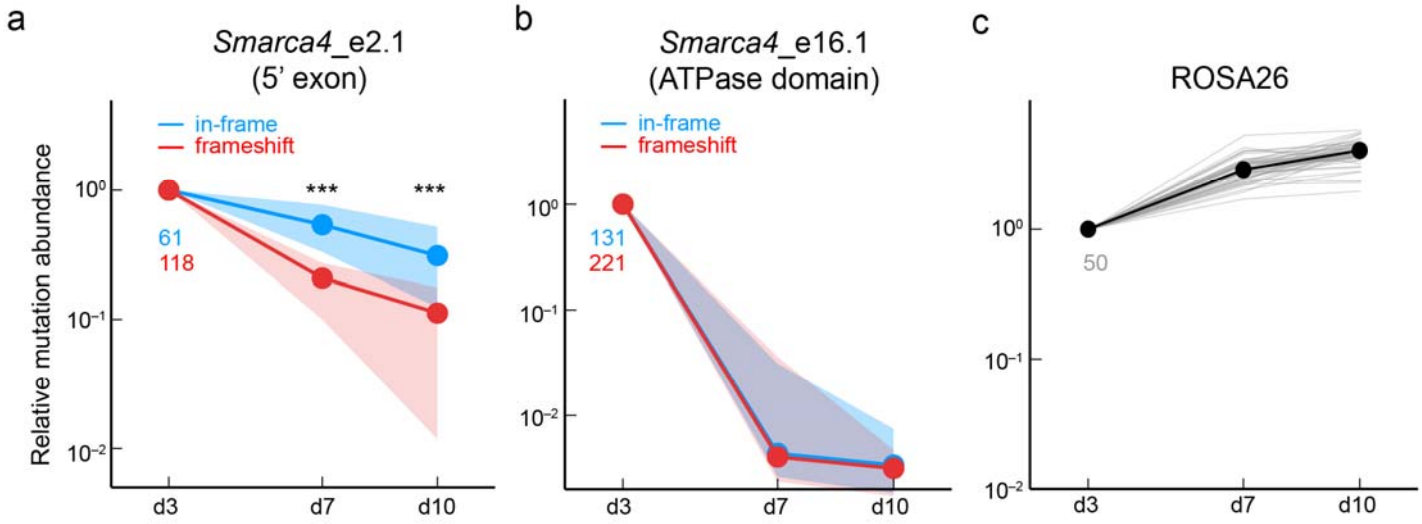
2 independent experiments is shown. (**b**) SURVEYOR assay of indel mutations induced by Brd4 sgRNAs at the indicated timepoints post-infection. Mutations induced by e3.3 undergo stronger negative selection than mutations induced by e3.1. (**c**) Deep sequencing-based analysis of CRISPR-mediated mutagenesis efficiency at the indicated Brd4 sgRNA cut sites performed at various timepoints post-infection. Illumina sequencing was used to quantify indel mutations at the corresponding to the sgRNA cut site. The GFP% at these timepoints was used to determine the overall indel% in transduced cells. ND: Not determined since the GFP % was low due to sever negative selection. (**d**) SURVEYOR assay of indel mutations induced by Smarca4 sgRNAs at the indicated timepoints post-infection. Mutations induced by e16.1 and e16.2 undergo stronger negative selection than mutations induced by e2.1 and e3.1. M: marker. (**e**) Analysis of CRISPR-mediated mutagenesis efficiency at the indicated Smarca4 sgRNA cut sites performed at various timepoints post-infection. Illumina sequencing was used to quantify indel mutations at the corresponding to the sgRNA cut site. The GFP% at these timepoints was used to determine the overall indel% in transduced cells. ND: Not determined since the GFP % was low due to severe negative selection.



Supplementary Figure 3

Brd4* BD1 sgRNAs do not exhibit off-target mutagenesis of homologous BD1 domains of *Brd2* and *Brd3.

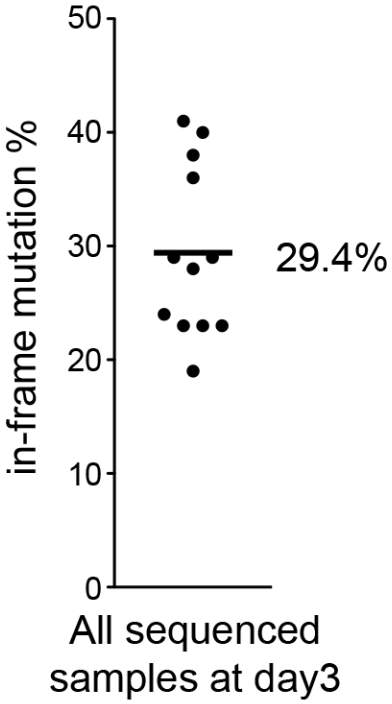
Analysis of CRISPR editing efficiency at the indicated BD1 domain-encoding exons of Brd2, Brd3, and Brd4, following transduction with Brd4 BD1 targeting sgRNAs e3.3 and e4.1. Analysis was performed at day 3 post-infection. The indel% was calculated based on the relative intensity of the DNA bands using ImageJ software. Results are representative of two independent biological replicates. M: marker. N.D.: not determined.



Supplementary Figure 4

Deep sequencing analysis of mutation abundance following CRISPR-targeting of different *Smarca4* or *Rosa26* regions.

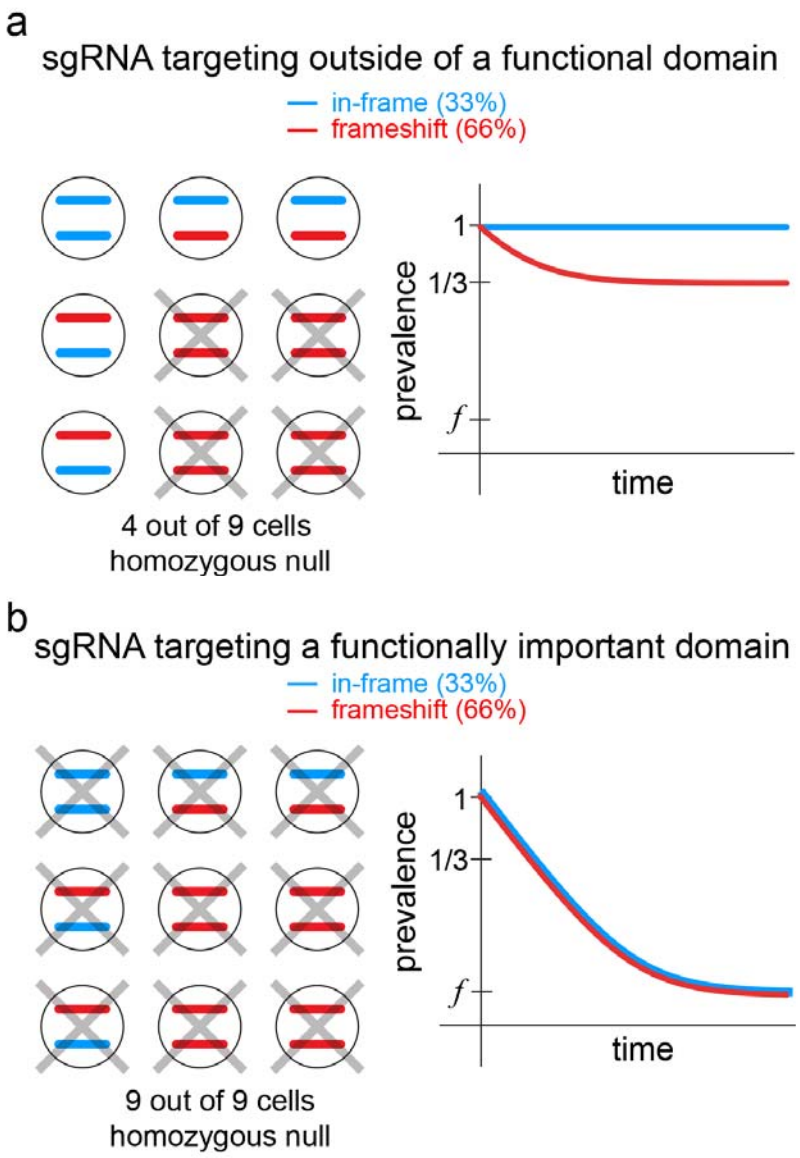
(**a-c**) This analysis was performed on PCR-amplified genomic DNA corresponding to the sgRNA cut site at the indicated timepoints. Indel mutations were categorized into two groups: in-frame (3n) or frameshift (3n+1, 3n+2). Nonsense mutations were also included with the frameshift category, however such mutations were rare in this analysis. Green and red numbers indicate the number of in-frame and frameshift mutants that were tracked, respectively. For a and b, dots of the same color indicate the median normalized abundance at the indicated time point for all mutations within each group; shaded regions indicate the interquartile range of normalized abundance values. For c, the relative abundance of 50 individual ROSA26 indels (indicated as light-gray lines) at indicated timepoints normalized to day 3 abundance. The black line represents the median normalized abundance across all 50 mutations. For a and b, significant differences between the enrichment values of the in-frame and frameshift mutations were assessed using a Mann-Whitney-Wilcoxon test; ** indicates $p < 0.01$, and *** indicates $p < 0.005$. The normalized abundance of each tracked mutation was defined as the ratio of the number of observed mutant sequences divided by the number of wild-type sequences, normalized by the value of this same quantity at day 3.



Supplementary Figure 5

Deep-sequencing analysis of in-frame mutation frequency induced by various sgRNAs.

Across 12 different sgRNAs used in this study, deep sequencing analysis of mutations at day 3 indicates an average frequency of in-frame mutations (3n) of 29.4%, with the remaining indel mutations being frameshifts, which matches well the expected ratio and the observations of others



Supplementary Figure 6

A model illustrating the expected genotypes and mutational abundance observed upon CRISPR targeting of different regions of an essential protein.

(**a, left**) Model for anticipated genotypes upon CRISPR mutagenesis of a 5' coding exon that lacks a functionally important domain, in which in-frame variants would retain functionality. If 33% of CRISPR mutations are in-frame and 66% are frameshift, then 4/9 of cells would be expected to have biallelic frameshift mutations, which would represent a homozygous null state. 5/9 of cells would carry at least one in-frame indel allele, which would retain functionality. This would render ~56% of cells in the population with a less severe phenotype. (**a, right**) The anticipated deep-sequencing based analysis of mutational abundance when targeting a 5' coding exon that lacks a functional domain. Since each in-frame mutation will cause the cell it resides in to be phenotypically unaffected, the prevalence of each in-frame mutation (relative to the wild type allele) will remain constant over time. Each frameshift mutation, on the other hand, has a 1/3 probability of being paired with an in-frame mutation and a 2/3 probability of being paired with another frameshift. Cells will be phenotypically affected more strongly in the latter case. Therefore, the prevalence of each frameshift mutation will first decrease then plateau at a value of 1/3. More precisely, the relative prevalence of in-frame (P_{if}) and frameshift (P_{fs}) mutations as a function of time will be

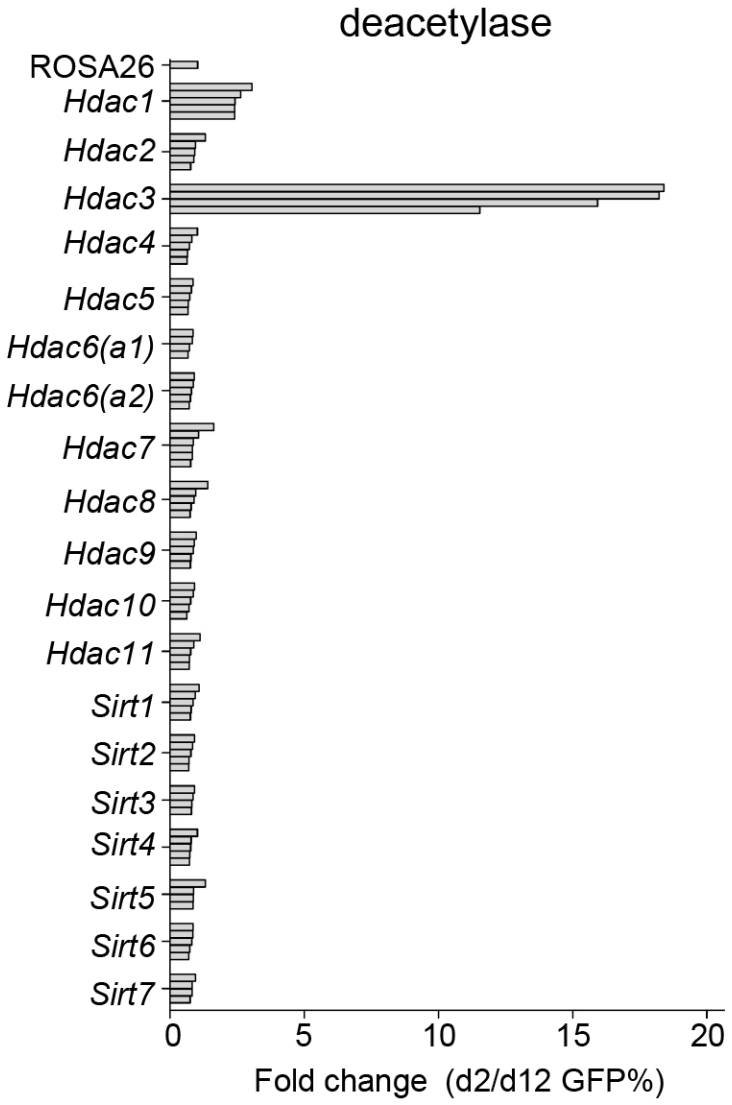
$$P_{if}(t) = 1, \quad P_{fs}(t) = \frac{1}{3} + \frac{2}{3}e^{-rt}.$$

The prevalence of both in-frame and frameshift mutation will decay at rate *r*.

(**b, left**) Model for anticipated genotypes upon CRISPR mutagenesis of an exon that encodes a functionally important domain, in which both in-frame and frameshift mutations will disable the protein. Nearly every cell in which both alleles are mutagenized will therefore lose the functionality of this protein and thus be phenotypically affected (**b, right**). The anticipated deep-sequencing based analysis of mutational abundance when targeting an exon

encoding a functionally important domain. The prevalence of both in-frame and frameshift mutation will decay at rate *r*. This decay will ultimately plateau at a value of *f*, where *f* is the failure rate of CRISPR mutagenesis, due to CRISPR either not mutagenizing both alleles within the cell or producing a non-disruptive mutation in the unobserved allele. Specifically,

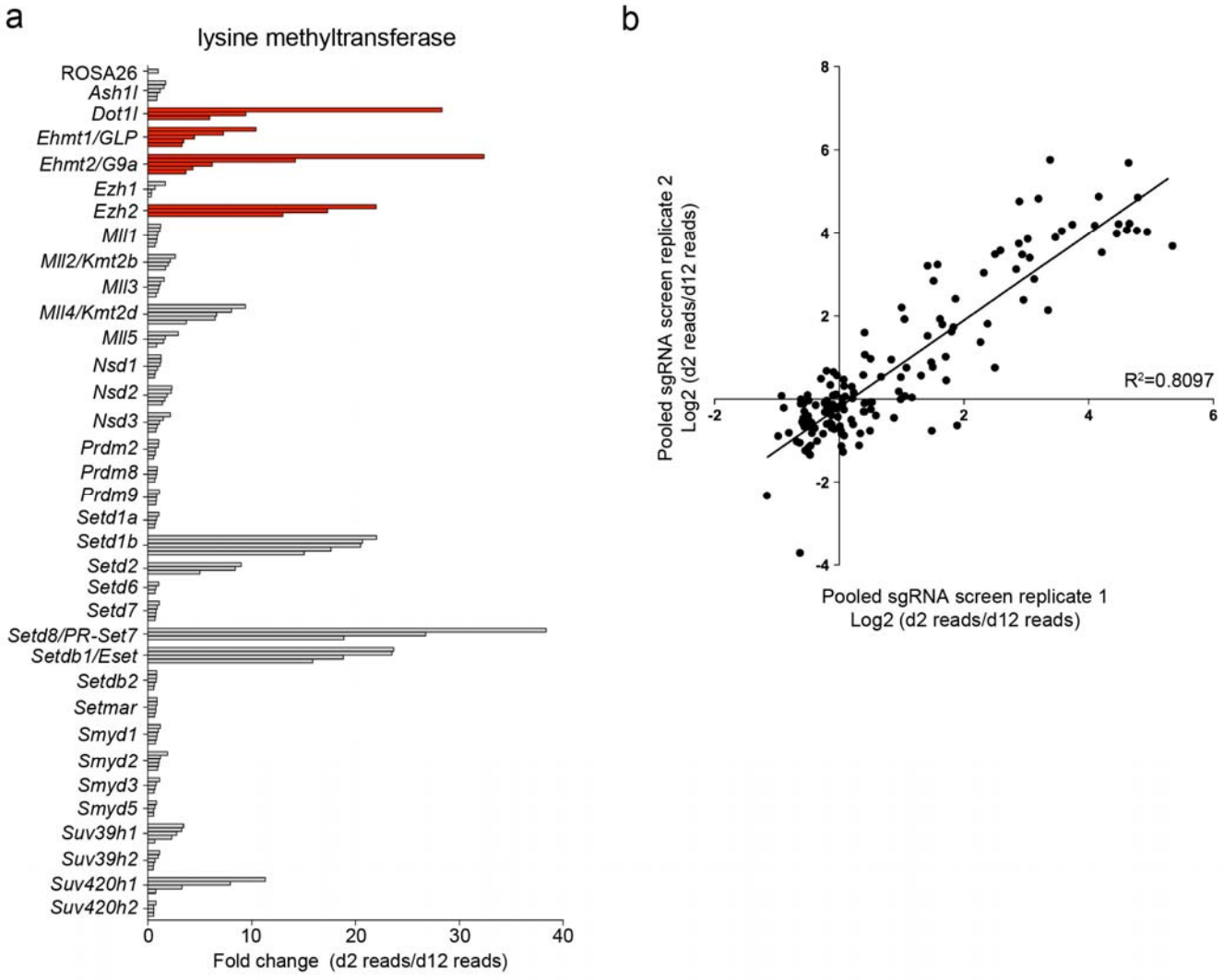
$$P_{1f}(t) = P_{fx}(t) = f + (1 - f)e^{-rt}.$$



Supplementary Figure 7

Deacetylase domain-focused CRISPR-Cas9 screen in murine MLL-AF9/Nras^{G12D} acute myeloid leukemia cells.

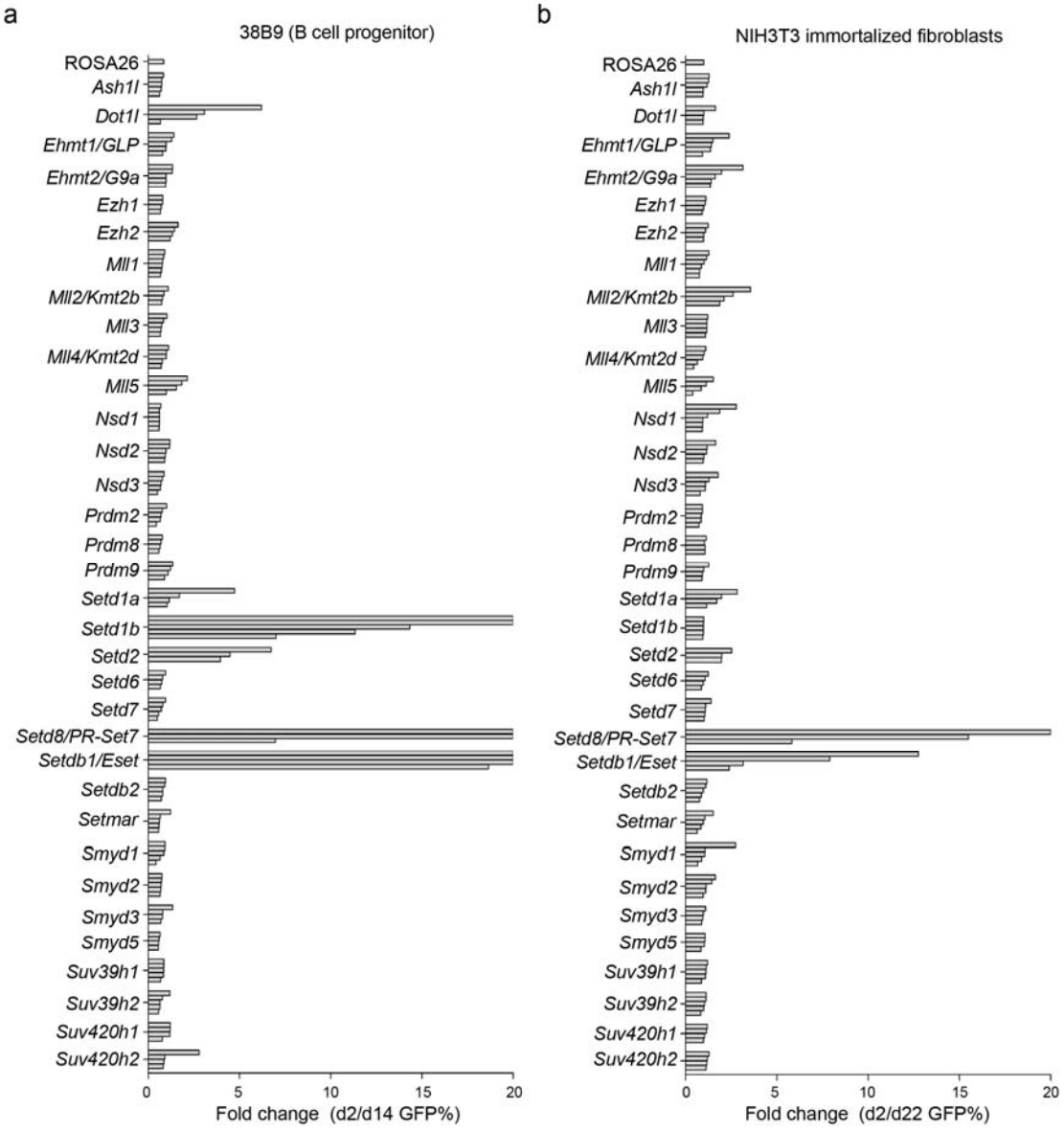
Summary of negative selection experiments with sgRNAs targeting the indicated domains plotted as fold change in GFP-positivity. Each bar represents the mean value of three independent biological replicates for an independent sgRNA targeting the indicated domain. The two deacetylase domains of HDAC6 are indicated as a1 and a2.



Supplementary Figure 8

Pooled sgRNA screen targeting lysine methyltransferase domains leads to similar results as analysis of individual sgRNAs using GFP reporters.

(**a**) Results of the pooled sgRNA screen evaluating lysine methyltransferase dependencies. The pooled library of sgRNAs was transduced into RN2c cells at a representation of ~500 transduced cells per sgRNA, followed by collection of genomic DNA at day 2 and day 12 post-infection. The sgRNA cassette was PCR-amplified from these samples and subjected to Illumina sequencing to measure the abundance of individual sgRNAs over time. The fold change in sgRNA abundance was calculated and plotted as the average of two independent biological replicates. Results were normalized to ROSA26 sgRNA. Red indicates the known drug targets within this class of regulators. The results closely match the findings obtained by scoring sgRNAs individually, shown in Figure 3. (**b**) Scatter plot that compares the fold change measurements between the two independent replicates.



Supplementary Figure 9

Lysine methyltransferase sgRNA screen performed in Cas9$^+$ 38B9 cells (murine B-cell progenitor line) and in Cas9$^+$ NIH3T3 cells (immortalized fibroblasts).

Cell lines were transduced with MSCV-Cas9-PGK-Puro followed by puromycin selection, prior to transduction with U6-sgRNA-EFS-GFP lentivirus. Summary of negative selection experiments with sgRNAs targeting the indicated domains plotted as fold change in GFP-positivity. A 20-fold cutoff was applied for visualization purposes.

mutagenesis efficiency

sgRNA	day3	day7	day10
Ezh2_e2.1	>95%	>95%	>95%
Ezh2_e19.2	>95%	>95%	N.D.
Dot1l_e1.1	>95%	>95%	>95%
Dot1l_e7.1	>95%	>95%	>95%
Dot1l_e11.2	>95%	>95%	N.D.

Supplementary Figure 10

Deep sequencing analysis of indel mutations induced by various *Ezh2* or *Dot1l* sgRNAs.

Analysis of CRISPR-mediated mutagenesis efficiency at the indicated either Ezh2 or Dot1l sgRNA cut sites performed at various timepoints post-infection. Ezh2_e2.1 and Dot1l_e1.1 sgRNAs target 5' coding exons. Ezh2_e19.2, Dot1l_e7.1, and Dot1l_e11.2 sgRNAs target methyltransferase domains. Illumina sequencing was used to quantify the CRISPR-induced indel mutations at the corresponding sgRNA cut site. The GFP% at these timepoints was used to determine the overall indel% in transduced cells. ND: Not determined since the GFP % was low due to sever negative selection.

Supplementary Discussion

Explanation for the apparent stronger negative selection of frameshift mutations when occurring in functionally important domains

In the deep sequencing analysis of CRISPR mutations shown in Figure 2i-k, we observed that frameshift mutations underwent negative selection when induced at any of the three sgRNA locations (BD1 or non-BD1 sites). However, the severity of negative selection is significantly less when targeting outside of BD1. The reasons for this are not immediately obvious, since it would be expected that truncating Brd4 at any of these 3 N-terminal sites should eliminate most of the full-length protein. However, it is important to consider the diploid nature of these cells. Each cell in the population will acquire a random CRISPR mutation on each copy of the *Brd4* gene. As depicted in Supplementary Figure 6, pairing of a frameshift mutation with an in-frame variant will likely prevent negative selection from occurring to the same severity as when a cell is homozygous for frameshift mutations. Hence, the functionality of in-frame variants will influence the negative selection behavior of frameshift mutations. Since in-frame mutations in a domain appear to lack functionality, it would be expected that frameshift mutations would more strongly deplete when targeted via CRISPR to a domain region.

Another potential explanation for these differential effects would be that different lengths of a truncated protein might retain varying levels of functionality or could potentially have differing degrees of dominant negative effects. It is also possible that varying levels of nonsense-mediated decay could influence the phenotypic consequences of these different frameshift mutations. It also a possibility that some of the frameshift mutations occurring at 5' exons could be 'rescued' by the use of an alternative start codon, which could restore expression of a nearly full-length protein. It is also worth emphasizing the findings in Supplementary Figure 5, where we observe a degree of variability in the frequency of in-frame mutations for certain sgRNAs. This reflects a degree of bias in the outcome of CRISPR mutagenesis, which can favor the formation of certain mutations. This variation in the frequency of in-frame mutations would also be expected to contribute to the variable severity of negative selection. Certain sgRNA sequence features may favor the formation of frameshift mutations[4, 5]. As a final consideration in this analysis, variation in the overall efficiency of CRISPR mutagenesis can also influence the ratios of the different genotypes.

Using deep sequencing-based measurements of mutation abundance to rule out off-target effects when validating dependencies identified from CRISPR screens.

We noted that the deep sequencing-based measurement of mutation abundance provided a useful means of excluding off-target effects, which has been a confounding variable in negative selection screens. Mutations induced by the *Brd4* sgRNA e3.1 exhibit a categorical separation of allele functionality for the in-frame (functional) and frameshift (non-functional) mutations (Fig. 2i). This pattern would not have occurred if negative selection was attributed due to mutagenesis of an off-target site, which would instead display a random pattern of negative selection when comparing frameshift and in-frame Brd4 variants. The consistency of this pattern across 75 distinct *Brd4* mutations provides strong evidence that the *Brd4* open reading frame encodes an essential protein in RN2c cells. Hence, performing a deep sequencing analysis of mutation abundance outside of critical domain can be useful for validating that a gene is essential.

Supplementary References

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